



REFERENCE

# The smysl Document Format

*Every construct of the .smy surface syntax, with a worked example for each*

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Vladimir Ulogov · 2026 · smysl 1.7.0 · format smysl/1.0 · kernel smysl.kernel/0.1

This reference covers one thing only: how to write a `.smy` document by hand — every record you can author, every field it takes, and the rules a validator will hold you to. For **why** the format is shaped this way, see `SMYSL_RATIONALE.typ`; this document only answers **how**. Every example below is real syntax, checked against the parser rather than invented for the page.

## THIS GUIDE IS NOT THE CONTRACT

It is a **writer's** reference — descriptive, and free to explain more than it obliges. What another implementation must obey to interoperate is `SMYSL_FORMAT_SPEC.md`, which is normative and deliberately a fraction of this length: identity and how a uid is derived, the deterministic-CBOR constraints, record framing, the round-trip fixed point, rule X and the conformance classes. Nothing else is required.

RFC SMYSL-1, which earlier versions of this guide referred to, is retired. It was the product idea rather than the doctrine, and the implementation outgrew it.

## How a document is put together

A `.smy` file is a flat sequence of **records**, each starting at column 0. Order does not matter to the meaning of the document — a claim can cite evidence that appears later in the file — though it is conventional to write things in the order a reader would want to meet them. Blank lines separate records but carry no meaning of their own.

## COMMENTS

A line beginning `#` or `//` at column 0 is a comment. Both markers, because an HJSON header inside a record already accepted both, and rejecting between records what was accepted within one made the surface contradict itself.

A comment is a comment **wherever it sits**, including inside a body. The alternative was worse: a body runs from the gist to the next record, so a comment between two records fell inside that range and became the previous unit's body, inventing content out of a note.

A body that genuinely needs to open a line with a marker escapes it — `\#` and `\\//`, and `\\` for a line that already starts with a backslash. Only those three, and only at column 0, so prose full of Windows paths and LaTeX needs no thought. The writer puts the escape back, so `fmt` stays a fixed point.

New in 0.6. Before it there was no escape at all: a Markdown heading or a line of C++ in a body was read as a comment and dropped in silence, which is a poor way to treat exactly the content this format carries.

No record carries a comment, so canonical form cannot reproduce one and `fmt` warns before dropping them. A note that has to survive belongs in a unit — typically `@question` or `@prose` — where it is content and travels like content.

Inside a header the markers work the same way, which has one consequence for values: a value that **begins** with `#` or `//` must be quoted, or the rest of the line — closing brace included — is read as a comment. The writer quotes such a value for you. A marker in the **middle** of a value is ordinary text, because a quoteless value runs to `,`, `}`, `]` or end of line without stopping at either marker. So `ref: grafana://board/12` needs no quotes and gets none.

New in 0.2. Before that there was no comment syntax at all, and such a line was `SMY-E001`. The quoting rule for values is 0.4: until then the writer emitted such a value bare and lost the record it belonged to.

Any other line that is not part of a record's header, gist, body, or detail is a diagnostic (`SMY-E001`), not an ignored remark.

Four kinds of record are ever hand-authored in surface syntax:

| CONSTRUCT              | WHAT IT DECLARES                                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <code>@doc</code>      | The document header — at most one per file, and optional.                                                                      |
| a unit                 | <code>@claim</code> , <code>@evidence</code> , and the other thirteen kernel types — the thing being claimed, cited, or asked. |
| <code>@rel</code>      | A typed edge between two units — <code>causes</code> , <code>rebutts</code> , <code>warrant</code> , and eleven more.          |
| <code>@thread</code>   | A named, ordered, role-annotated walk over units — a brief, a narrative, an analysis.                                          |
| <code>@schema</code>   | A declaration of an extension: the unit types and relation kinds it adds. Rarely needed, and one line when it is.              |
| <code>@withdraw</code> | An edge that should no longer be followed, and who withdrew it. Since 1.4.                                                     |
| <code>@resolve</code>  | A record that a disagreement was reviewed, and by whom. Since 1.4.                                                             |

Other record kinds exist in the format and none is something you write directly. An **attestation** is stamped on automatically by whatever tool ingests or authors a unit; a **contention** is what `merge`

produces when two documents disagree; a **pack manifest** is **pack**'s receipt; a **schema declaration** names an extension a store depends on.

A **label binding** is the one worth knowing about, because it explains a number. Writing `@claim c/cache-cold` declares both a unit and a name for it, and the name travels as its own record — so a labelled unit yields two. That is why **check** reports more records than you typed, and why **units** rather than **records** is the figure to read when you want to know how much document you have.

The binding has to be separate because a label is **not identity**: it is not hashed, and renaming one must not produce a different unit. New in 0.2 — before it, labels survived a parse and not a store round trip, so a document that had been through **merge** came back with every reference spelled as a bare uid.

## The `@doc` header

If present, `@doc` must be the first thing in the file, and it fixes the document's own identity and the terms under which everything else in it should be read:

```
@doc smysl/0.1 {
  id: v/f1
  intent: incident-brief
  lang: en
  requires: ["smysl.kernel/0.1"]
  granularity: { profile: default }
  roots: [f/root-cause]
}
```

| FIELD              | DEFAULT                | MEANING                                                                                                 |
|--------------------|------------------------|---------------------------------------------------------------------------------------------------------|
| <b>id</b>          | <b>v/doc</b>           | This document's own identifier — a view id, <b>v/&lt;slug&gt;</b> .                                     |
| <b>intent</b>      | empty                  | A free-text label for what the document is for, e.g. <b>incident-brief</b> .                            |
| <b>lang</b>        | <b>en</b>              | A BCP-47 language tag — up to 35 characters, hyphen-separated alphanumeric segments.                    |
| <b>requires</b>    | empty                  | Schemas a full-fidelity reader must implement: the kernel schema itself and any extensions.             |
| <b>roots</b>       | empty                  | The units this view is <b>about</b> — everything reachable from them is in the view; nothing is copied. |
| <b>threads</b>     | empty                  | Thread ids this view publishes, if any.                                                                 |
| <b>granularity</b> | <b>default</b> profile | How much one unit is allowed to say — see below.                                                        |

A view is not a container: **roots** names starting points, and everything reachable from them via **deps**, **grounds**, and relation edges belongs to the view at zero copying cost. That is also what makes merging two documents a plain union rather than a conflict.

## Granularity

**granularity** bounds how much a single unit is allowed to say, as three presets or a custom mix. Granularity constrains **production** — what you should write — not the store: a merged corpus mixing granularities is legal, not an error.

| PROFILE | GIST (L0)   | BODY (L1)        | ADMISSION                              |
|---------|-------------|------------------|----------------------------------------|
| coarse  | ≤ 30 tokens | 120 – 400 tokens | topical — one unit may bundle a topic  |
| default | ≤ 30 tokens | 40 – 120 tokens  | single-assertion — one unit, one claim |
| fine    | ≤ 30 tokens | 20 – 60 tokens   | single-assertion                       |

Under **single-assertion** admission — the default — a body that bundles more than one claim into a single unit is **SMY-E040**, an error, not a style complaint: split it into two units and a relation between them instead. A body outside its profile's range is only **SMY-W041**, a warning.

## Units — the shared anatomy

Every unit, whatever its type, shares the same header and the same three-level text. Here is the fullest ordinary example the format has:

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait], salience:
0.8 }
~ The eu-west connection pool is saturated.
  Waits climbed from a steady 2 ms baseline to over 300 ms in the same hour the 4.2
  rollout reached eu-west, and no other shard's pool moved.

--
Per-shard breakdown at hourly resolution: eu-west crosses 250 ms at 09:14 UTC,
forty
minutes after the rollout's first canary step there. No other shard's pool exceeds
its usual 5 ms band over the same seven-day window.
```

| FIELD        | REQUIRED WHEN                                                    | WHAT IT IS                                                                                                                                                                                                                                                                        |
|--------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| label        | optional                                                         | The word right after the type, before <code>{</code> , e.g. <code>c/pool-saturation</code> — how the rest of the file refers to this unit. Not identity: two labels never make two units, and a unit needs no label at all if nothing else in the file must reference it by name. |
| status       | always — defaults to <b>speculative</b>                          | How sure this unit is entitled to be — the six-rung ladder, next section.                                                                                                                                                                                                         |
| gist (~ ...) | always                                                           | The one required line of text — L0. Must sit on the line immediately after the header, with no blank line between.                                                                                                                                                                |
| body         | needed by <b>derived</b> / <b>inferred</b> gists that lean on it | L1 — a paragraph or two, the ordinary prose account.                                                                                                                                                                                                                              |
| detail       | only after a body                                                | L2 — unbounded, introduced by a lone <code>--</code> line. A detail with no body above it is <b>SMY-E023</b> .                                                                                                                                                                    |
| deps         | when the gist needs context to parse                             | Other units this one's <b>wording</b> depends on to be understood — not evidence, just prerequisite reading.                                                                                                                                                                      |

| FIELD           | REQUIRED WHEN                                | WHAT IT IS                                                                                                                     |
|-----------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>grounds</b>  | required by <b>derived</b> / <b>inferred</b> | Other units this one's <b>truth</b> rests on. Retract one and this unit's status is re-opened.                                 |
| <b>source</b>   | required by <b>measured</b> / <b>cited</b>   | Where this unit grounds out externally – see below.                                                                            |
| <b>salience</b> | optional                                     | An authored override in <b>[0,1]</b> , clamped and quantised to 1/1024. Left unset, salience is computed rather than declared. |

The gist must immediately follow the header – not even a blank line may separate them (SMY-E021). A gist continuation line is indented by exactly two spaces; a body or detail line is not indented at all. The assembled gist is trimmed: whitespace around it is never content, because the reader strips the ~ sigil and the space after it, so a gist that carried a leading space could not be written back and read again unchanged.

#### WHAT IS NOT CONTENT

Identity is content, so anything the format treats as **not** content has to be settled before a uid is computed – otherwise two documents that say the same thing hash differently. Three things are normalised on the way in:

- **Line endings.** A CRLF file and an LF file are the same document. Every carriage return at the end of a line is stripped, not just one. A carriage return **inside** a line is ordinary text and survives.
- **Whitespace around a gist,** as above.
- **Unicode form.** All text is normalised to NFC – including the unknown header keys and values that rule X carries verbatim. é written as one code point and é written as e plus a combining accent are the same text and get the same uid.
- **A repeated field.** Writing **deps** twice in one header keeps the first and discards the rest. Before 0.5 the second was carried as an unknown key under rule X and written back out as a plain **deps:** line, which the next parse then read as the field – so the document lost content by being reformatted. There is no way to spell a second **deps** in surface syntax, so first-wins is the only rule that round-trips.

Each of these was a real defect before 0.4, and the Unicode one was the worst of them: it failed silently in release builds, so two peers could write identical content and disagree about its identity.

#### Status: the six-rung ladder

| STATUS             | NEEDS          | MEANING                                                             |
|--------------------|----------------|---------------------------------------------------------------------|
| <b>unfounded</b>   | –              | Reachable only by retraction; MUST NOT be authored (SMY-E034).      |
| <b>speculative</b> | neither        | Offered, not yet grounded – the default if you omit <b>status</b> . |
| <b>inferred</b>    | <b>grounds</b> | A model reasoned it out from other units.                           |
| <b>derived</b>     | <b>grounds</b> | Computed from other units by a deterministic procedure.             |
| <b>cited</b>       | <b>source</b>  | A source you can open says so.                                      |
| <b>measured</b>    | <b>source</b>  | An instrument recorded it.                                          |

Two mechanical rules sit on top of this table. **Rule M**: a unit's status can never outrank the weakest thing in its own `grounds` (SMY-E030) — a `derived` claim resting on a `speculative` one is itself no better than speculative, whatever status you write. **Rule T**: the `rung` an agent is working from caps what status it may even attempt:

| RUNG                  | CEILING               | WHAT IT IS                                                                        |
|-----------------------|-----------------------|-----------------------------------------------------------------------------------|
| <code>computed</code> | <code>derived</code>  | A deterministic tool, calculation, or parser.                                     |
| <code>document</code> | <code>cited</code>    | A user-supplied document or dataset.                                              |
| <code>web</code>      | <code>cited</code>    | Fetches content, gated.                                                           |
| <code>model</code>    | <code>inferred</code> | The model's own parametric knowledge — never higher, however confidently phrased. |

No rung reaches `measured` on its own, and the way it is reached is worth stating exactly, because the obvious reading is wrong. It takes an attestation recording `op: imported at the computed rung` — a deterministic tool transcribing a reading. The `op` alone is not enough: `ingest` also records `imported`, because it too transcribes rather than authors, so unlocking on the `op` by itself would let a model mark anything it read in a document as measured. Ingest runs at `document`, `web` or `model` and stays capped there.

#### WHY YOU CAN STILL TYPE IT

You may write `status: measured` in a `.smy` file and `check` will pass it, which looks like a contradiction and is not. Rule T is checked against a unit's **attestation**, and a `.smy` file cannot express one — provenance has no surface syntax. With nothing recorded about where a unit came from, there is no ceiling to check it against, so the rule stands aside rather than guessing. What you have written is an unchecked claim, not a licensed one, and it becomes checkable the moment the unit acquires provenance.

The producer that writes `measured` **with** the provenance that licenses it is the `import` command, which transcribes a table of readings. A unit whose attestation records some other `op` is SMY-W035.

### Keys the grammar does not know

A header key that is not one of the fields above is **not** an error. It is kept, verbatim, in the unit's payload and written back out in the same place — the surface half of rule X. A reader from a later version, or from a team that records something yours does not, loses nothing by passing through yours.

```
@data t/latency { status: measured, source: { kind: file, ref: "by-region.csv" },
  x.stats/method: "empirical-quantile",
  columns: ["region", "p50_ms", "p95_ms"], rows: 6 }
~ Checkout latency by region, June 2026.
```

`rows`, `columns` and `x.stats/method` are none of the kernel's business. They survive a parse, a round trip through the binary encoding, and a merge — and they come back in canonical order rather than the order you typed them, which is why `smyfmt` returns the unit above with `rows` first and `x.stats/method` last. Keys sort by encoded length, then content. That is not tidiness: identity is computed from the encoded bytes, so two people who record the same thing in a different order still compute the same unit.

## TWO NAMES THE TOOLING WRITES

Ingest puts two of its own keys in the same place, and you will meet them reading a staged file rather than writing one. `ingest:quote` carries the span of the source document a unit was drawn from — checked against that document, so a quote that is not in it is an error rather than a decoration. `ingest:unrepaired` marks a span that survived its repair budget and was kept as opaque prose rather than dropped. Neither is reserved by the grammar; both are conventions you can read, write, or ignore.

## Source references

```
source: { kind: metric, ref: "pool.wait_ms{shard=eu-west}", captured: 2026-07-09 }
```

`kind` is one of five, `ref` (or the longer spelling `reference`) is a free-text pointer whose shape depends on the kind, and `captured` is an optional `YYYY-MM-DD` date.

| KIND                | TYPICAL REF                    | CAPTURE DATE                                                        |
|---------------------|--------------------------------|---------------------------------------------------------------------|
| <code>url</code>    | a web address                  | <b>Required</b> — a fetched page is unverifiable later without one. |
| <code>file</code>   | a path or filename             | optional                                                            |
| <code>metric</code> | a metric name or query         | optional                                                            |
| <code>tool</code>   | a tool or command name         | optional                                                            |
| <code>doc</code>    | a document reference or anchor | optional                                                            |

## The thirteen unit types

`KernelType` actually enumerates fifteen names, but two of them — `contention` and `packinfo` — name records that tooling produces (`merge`, `pack`); nothing below shows them being hand-authored, because in ordinary use they never are. The other thirteen are yours to write. Each example is complete and independently valid.

| TYPE                      | WHAT IT RECORDS                                                    |
|---------------------------|--------------------------------------------------------------------|
| <code>claim</code>        | An assertion about the world.                                      |
| <code>evidence</code>     | A reading, instrument, or observation offered to ground a claim.   |
| <code>definition</code>   | What a term means, fixed so later claims can rely on it.           |
| <code>question</code>     | An open prompt the corpus has not yet answered.                    |
| <code>hypothesis</code>   | A candidate answer, offered before it is tested.                   |
| <code>finding</code>      | A conclusion resting on other claims.                              |
| <code>procedure</code>    | A course of action — a “what to do,” not a “what is.”              |
| <code>decision</code>     | A commitment a person or team made.                                |
| <code>constraint</code>   | A rule the rest of the reasoning has to respect.                   |
| <code>observation</code>  | A plain record of what was seen.                                   |
| <code>data</code>         | A pointer to a dataset, rather than one reading.                   |
| <code>artifact-ref</code> | A pointer to something outside the document — a dashboard, a repo. |

| TYPE  | WHAT IT RECORDS                                                                     |
|-------|-------------------------------------------------------------------------------------|
| prose | Free-standing text that fits none of the above — still a first-class, citable unit. |

```
@evidence e/pool-wait { status: measured, source: { kind: metric, ref:
"pool.wait_ms{shard=eu-west}", captured: 2026-07-09 } }
~ Pool acquisition wait rose from 2 ms to 310 ms over the same window.
```

An instrument reading grounds out externally, so **measured** demands a **source** — here a metric reference with the date it was captured.

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait] }
~ The eu-west connection pool is saturated.
```

A model reasoned this from the evidence above; **inferred** demands **grounds**, and rule M means this claim can never outrank **e/pool-wait** even if the wording sounds certain.

```
@definition d/p95 { status: cited, source: { kind: doc, ref: "sre-
handbook#latency" } }
~ The 95th percentile of request latency over a one-minute window.
```

A definition is grounded in a document a reader can open, not the author’s own say-so — without it, “tripled” in a later claim would have no fixed meaning.

```
@question q/root-cause { status: speculative }
~ What actually caused the eu-west latency spike?
```

A question needs neither **source** nor **grounds** — **speculative** is the floor, an open prompt nothing has answered yet.

```
@hypothesis h/pool-exhaustion { status: speculative }
~ Connection-pool exhaustion under the 4.2 rollout is driving the p95 rise.
```

A candidate answer to the question above, offered before it is tested against evidence.

```
@finding f/root-cause { status: inferred, grounds: [c/pool-saturation, c/
regression] }
~ Pool saturation is the leading cause but is not consistent with the canary.
```

A finding rests on other claims rather than on raw evidence; rule M caps it at the weakest of the two grounds it names.

```
@procedure p/rollback { status: speculative }
~ Roll the eu-west pool config back to the 4.1 settings and re-run the canary.
```

A course of action, not a fact — still a checkable unit, just one whose content is a “what to do.”

```
@decision de/rollback-approved { status: inferred, grounds: [f/root-cause] }
~ Approved: roll back the eu-west pool config tonight.
```



A commitment, grounded in whatever finding justified it — retract the finding and this decision’s grounding is reopened along with it.

```
@constraint co/no-friday-deploys { status: cited, source: { kind: doc, ref: "ops-
runbook#change-freeze" } }
~ No production rollbacks after 16:00 on a Friday.
```

A rule the rest of the reasoning has to respect, sourced from wherever that rule is actually written down.

```
@observation ob/canary-quiet { status: measured, source: { kind: metric, ref:
"canary.p95" } }
~ The 4.2 canary showed no latency regression over the same window.
```

A plain record of what was seen — it doesn’t ground anything by itself yet, but it can be cited later, exactly like any other unit.

```
@data da/latency-series { status: measured, source: { kind: file, ref: "auth-
p95-2026-07.csv" } }
~ Hourly p95 auth latency, all shards, 2026-07-02 through 2026-07-09.
```

Points at a dataset rather than asserting a single reading — the source is the file itself.

```
@artifact-ref ar/dashboard { status: cited, source: { kind: url, ref: "https://
grafana.internal/d/api-latency", captured: 2026-07-09 } }
~ The latency dashboard the on-call team watches.
```

Points at something that exists outside the document. Its source kind is `url`, the one kind where `captured` is mandatory rather than optional.

```
@prose pr/note { status: speculative }
~ Worth a follow-up: check whether the 4.1 pool size was ever tuned for eu-west's
traffic shape.
```

The escape hatch — a citable aside that doesn’t fit any more specific type.

#### NAMESPACE CONVENTION, NOT GRAMMAR

`e/`, `c/`, `d/`, `f/`, and the rest above are a **habit**, not a rule. A label is any `ident/ident` pair — lowercase letters, digits, `-`, `_` — and nothing in the grammar reserves any prefix for any unit type. Pick short, memorable prefixes and stay consistent within a file; the parser does not care what they are.

## Relations (`@rel`)

A relation is a typed, directed edge between two units, written on one line:

```
<from> --<kind>--> <to> [{ weight: <0..1>, note: <ref> }]
```

```
@rel c/canary-clean --rebuts--> c/pool-saturation { weight: 0.6 }
```

A direct objection. `rebuts` is the one kind rule R pins: wherever `c/pool-saturation` survives into a packed or rendered view, this edge — and the unit it points from — travels with it. `weight` is an optional strength for the rebuttal itself, clamped to `[0,1]`.

```
@rel c/pool-saturation --causes--> c/regression
```

Says the saturation claim is what produced the regression. `causes` is one of three **ordering** kinds (with `enables` and `sequences`) that a thread’s automatic ordering can sort by, and one of two **support-bearing** kinds (with `answers`) that carry salience rank forward through the graph.

```
@rel d/p95 --warrant--> c/regression
```

Says the definition licenses reading the claim the way it is read — without it, “tripled” has no fixed meaning to check.

```
@rel c/pool-saturation-v2 --supersedes--> c/pool-saturation
```

Marks the newer claim as replacing the older one outright. `supersedes` and `retracts` are the two **lifecycle** kinds — the only ones that change what the corpus should currently be read as believing, rather than adding a new relationship alongside what was already there.

```
@rel c/pool-saturation --x.sre/mitigates--> p/rollback
```

An unrecognised kind — anything shaped `x.<domain>/<kind>` — is kept and stays routable rather than dropped (SMY-W013); a reader without the `x.sre` extension treats it as a plain `elaborates` edge instead of losing it.

The warning goes away when the file declares the kind it uses:

```
@schema x.sre/v1 { version: 1, relations: [x.sre/mitigates] }
```

The declaration travels with the store through `merge` and `fmt`. A misspelled key in it is an error rather than something skipped — a `relation:` quietly ignored would leave the kind warning under a file that appears to declare it — and declaring a kernel kind such as `causes` is SMY-E012, because an extension may add but never redefine.

An edge can be **withdrawn** when a reviewer finds the edge itself wrong — two statements a model matched as opposed that are not — without retracting either unit:

```
@withdraw c/canary-clean --rebuts--> c/pool-saturation { agent: human:vladimir, ts: [1726500000000, 0], reason: c/not-opposed }
```

The `@rel` line stays and the edge is no longer followed: a withdrawn rebuttal no longer travels with its claim. `retracts` and `supersedes` edges cannot be withdrawn. When a disagreement is real and both sides stand, the review is recorded instead, naming the edge or the contention merge reported:

```
@resolve k/ccm3actwjtti65famnoe6mapo5d { agent: human:vladimir, ts: [1726500000000, 0], note: c/both-stand }
```

A resolution decides nothing — the rebuttal still travels with the claim — and takes the item off the list `mysl review` prints. `ts` is milliseconds and a counter, as for `@thread`. Both words are reserved since 1.4, so an older `mysl` rejects a file that uses them.

The fourteen kernel kinds, for reference:

| KIND                     | READS AS                                                                          |
|--------------------------|-----------------------------------------------------------------------------------|
| <code>elaborates</code>  | adds detail without changing the claim                                            |
| <code>contrasts</code>   | sets two things against each other                                                |
| <code>concedes</code>    | grants a point while maintaining the larger claim                                 |
| <code>causes</code>      | one thing produced another ( <b>ordering, support-bearing</b> )                   |
| <code>enables</code>     | one thing made another possible without directly producing it ( <b>ordering</b> ) |
| <code>exemplifies</code> | gives a concrete instance of a general claim                                      |
| <code>conditions</code>  | states an “only if” dependency                                                    |
| <code>sequences</code>   | orders two things in time, with no causal claim ( <b>ordering</b> )               |
| <code>answers</code>     | resolves a question ( <b>support-bearing</b> )                                    |
| <code>rebutts</code>     | directly objects to the target — pinned by rule R                                 |
| <code>warrant</code>     | licenses an inference or a reading                                                |
| <code>backs</code>       | supports a warrant                                                                |
| <code>supersedes</code>  | replaces the target outright ( <b>lifecycle</b> )                                 |
| <code>retracts</code>    | withdraws the target outright ( <b>lifecycle</b> )                                |

## Threads (`@thread`)

A thread is a named, ordered, role-annotated walk over units — the same units the corpus already has, arranged into a reading order for one audience. Its header takes a `schema` (which fixes the allowed roles), an `owner` (an agent id — a thread is **owned** state, last- writer-wins **per owner**, so two agents publishing the same thread id never conflict), and a gist. Steps follow, indented by two spaces:

```
<role> → <unit>[: <note>]
```

Each of the five schemas fixes a different sequence of roles. One worked example per schema:

```
@thread t/brief { schema: brief, owner: "human:vladimir" }
~ Auth p95 tripled in eu-west; pool saturation is leading but contested.
  bottom-line → f/root-cause
  support → c/pool-saturation
  risk → c/canary-clean
```

`brief` — `bottom-line`, `support`, `risk`, `ask` — walks a reader from the one-line verdict down to what backs it and what argues against it. Not every role has to be used.

```
@thread t/root-cause-qa { schema: qa, owner: "human:vladimir" }
~ What caused the latency spike, and how sure are we?
  question → q/root-cause
  evidence → e/pool-wait
```

```
answer → f/root-cause
caveat → c/canary-clean
```

**qa** — **question, evidence, answer, caveat** — is literally a question answered in public, with its evidence and its own best objection attached.

```
@thread t/incident-story { schema: narrative, owner: "human:vladimir" }
~ How a quiet Tuesday became an eu-west incident.
  setup → ob/canary-quiet
  complication → e/pool-wait
  turn → c/pool-saturation
  resolution → de/rollback-approved
  coda → f/root-cause
```

**narrative** — **setup, complication, turn, resolution, coda** — orders the same kind of units as a story, for an audience that needs to follow how the conclusion was reached.

```
@thread t/deep-dive { schema: analysis, owner: "model:anthropic/claude-opus-5" }
~ Full trace of the eu-west pool-saturation analysis.
  context → ob/canary-quiet
  tension → e/pool-wait
  approach → h/pool-exhaustion
  finding → c/pool-saturation
  rebuttal → c/canary-clean
  implication → f/root-cause
  next → p/rollback
```

**analysis** — **context, tension, approach, finding, rebuttal, implication, next** — is the longest schema, a full research trace with its own slot for what to do next.

```
@thread t/rollback-plan { schema: plan, owner: "human:vladimir" }
~ Rolling eu-west back to 4.1 tonight.
  goal → de/rollback-approved
  constraint → co/no-friday-deploys
  step → p/rollback
  risk → c/canary-clean
```

**plan** — **goal, constraint, step, decision, risk** — orders the units that make up a course of action rather than an argument.

#### ARROWS AND NOTES

Either → (U+2192) or the ASCII -> is accepted when **reading**; the canonical writer (smysl fmt) always emits →. A step may carry a trailing : note after the reference, as free text — support → c/pool-saturation: the strongest single reading.

**risk** appears in both **brief** and **plan** — it is one role shared between schemas, not two separate ones. A step whose role the thread's schema does not declare is not rejected; it is reported as a **foreign role**, kept rather than dropped.

## Identifiers, at a glance

| KIND                          | SHAPE                                         | EXAMPLE                                                                     |
|-------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|
| Label                         | <code>ident/ident</code> , lowercase          | <code>c/pool-saturation</code>                                              |
| Thread / view / contention id | same shape, label-typed                       | <code>t/brief</code> ,<br><code>v/f1</code> , <code>k/pool-vs-index</code>  |
| Agent id                      | <code>(model\ human\ tool):name[/tag]</code>  | <code>model:anthropic/claude-opus-5</code> ,<br><code>human:vladimir</code> |
| Kernel type                   | a bare word                                   | <code>claim</code> , <code>evidence</code>                                  |
| Kernel schema                 | <code>smysl.kernel/MAJOR[.MINOR]</code>       | <code>smysl.kernel/0.1</code>                                               |
| Extension schema              | <code>x.&lt;domain&gt;/&lt;segment&gt;</code> | <code>x.sre/incident</code>                                                 |
| Language tag                  | BCP-47-shaped, ≤ 35 chars                     | <code>en</code> , <code>en-GB</code> , <code>zh-Hans-CN</code>              |
| Content hash (uid)            | <code>b3:</code> + base32, machine-facing     | <code>b3:cvhirtgs2mpvli2ethhyeo32uf...</code>                               |

You will essentially never type a `b3:` hash by hand — that is exactly what labels are for. A hash is what a unit’s identity actually is; a label is the name you gave it so you never have to write the hash yourself. An agent id’s kind is fixed to one of three words; a model id’s segment after the first `/` groups corroboration by provider, which is why `model:anthropic/claude-opus-5` and `model:anthropic/claude-haiku-4-5` still corroborate as the same provider.

## Validation, at a glance

A parse never simply fails: a malformed record becomes a diagnostic with a byte span, and parsing recovers at the next record start, so one bad unit costs you that unit, not the whole file. The codes below are the ones ordinary hand-authoring runs into.

| CODE     | WHAT IT MEANS                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------|
| SMY-E001 | General surface parse error — a malformed header, an unknown type, a stray line.                          |
| SMY-E020 | Body or detail names a unit’s identifier that isn’t listed in <code>deps</code> or <code>grounds</code> . |
| SMY-E021 | Missing gist, or the gist doesn’t immediately follow the header.                                          |
| SMY-E022 | Gist exceeds the profile’s <code>l0_max</code> .                                                          |
| SMY-E023 | <code>detail</code> present without a <code>body</code> above it.                                         |
| SMY-E030 | Rule M — status exceeds the weakest of its own grounds.                                                   |
| SMY-E031 | <code>derived/inferred</code> with empty <code>grounds</code> .                                           |
| SMY-E032 | <code>measured/cited</code> without a <code>source</code> .                                               |
| SMY-E033 | Rule T — status exceeds the ceiling for the authoring rung.                                               |
| SMY-E034 | <code>unfounded</code> authored directly — only reachable by retraction.                                  |
| SMY-E040 | More than one assertion in a body under <code>single-assertion</code> admission.                          |
| SMY-E060 | A label referenced but never defined anywhere in the document.                                            |
| SMY-E061 | A cycle in <code>deps</code> or <code>grounds</code> .                                                    |

| CODE     | WHAT IT MEANS                                                                 |
|----------|-------------------------------------------------------------------------------|
| SMY-W013 | Unknown relation kind — kept, treated as <code>elaborates</code> for closure. |
| SMY-W035 | <code>measured</code> on an authored (not imported) unit.                     |
| SMY-W041 | Body length outside the profile's <code>ll_range</code> — advisory only.      |

Three more you will only meet in a document a model produced, never one you typed. They are listed because you will read them in a staged file and should know they are not your mistake:

| CODE     | WHAT IT MEANS                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMY-W036 | Rule M applied at ingest — a unit claimed more than its grounds support and was lowered to what they do. The record of a model overreaching, not work outstanding. |
| SMY-E307 | An attributed <code>ingest:quote</code> does not occur in the source document. A fabricated attribution, and the one thing here worth another turn of the model.   |
| SMY-W308 | The quote occurs only loosely — elided or reworded. Honest attribution with a clause dropped.                                                                      |

#### AT A GLANCE

- Four constructs are hand-authored: `@doc` once, units as many as you need, `@rel` for typed edges, `@thread` to walk them in an order.
- Every unit shares one anatomy — label, status, gist/body/detail, `deps/grounds`, `source`, `salience` — whatever its type.
- Status only ever falls, never rises, across `grounds` (rule M) and is ceilinged by the rung doing the authoring (rule T); `measured` needs `op: imported` at the `computed` rung, which in practice means `smysl import` and nothing else.
- Thirteen unit types are yours to write; `contention` and `packinfo` are tooling-only.
- Fourteen relation kinds, five thread schemas, twenty-four roles — all closed sets, all with an escape hatch (`x.<domain>/<kind>`) that degrades gracefully rather than being dropped.
- Header keys the grammar does not know are kept verbatim in the payload rather than refused, so a document from a later version or another team passes through yours intact.
- A `#` or `//` line at column 0 is a comment, and no record carries one — `fmt` warns before dropping them. There is still no blank line permitted between a header and its gist, and no reserved label namespaces: only what the grammar actually checks.
- A labelled unit yields two records, because the label travels as its own binding rather than inside the hashed content that decides identity.